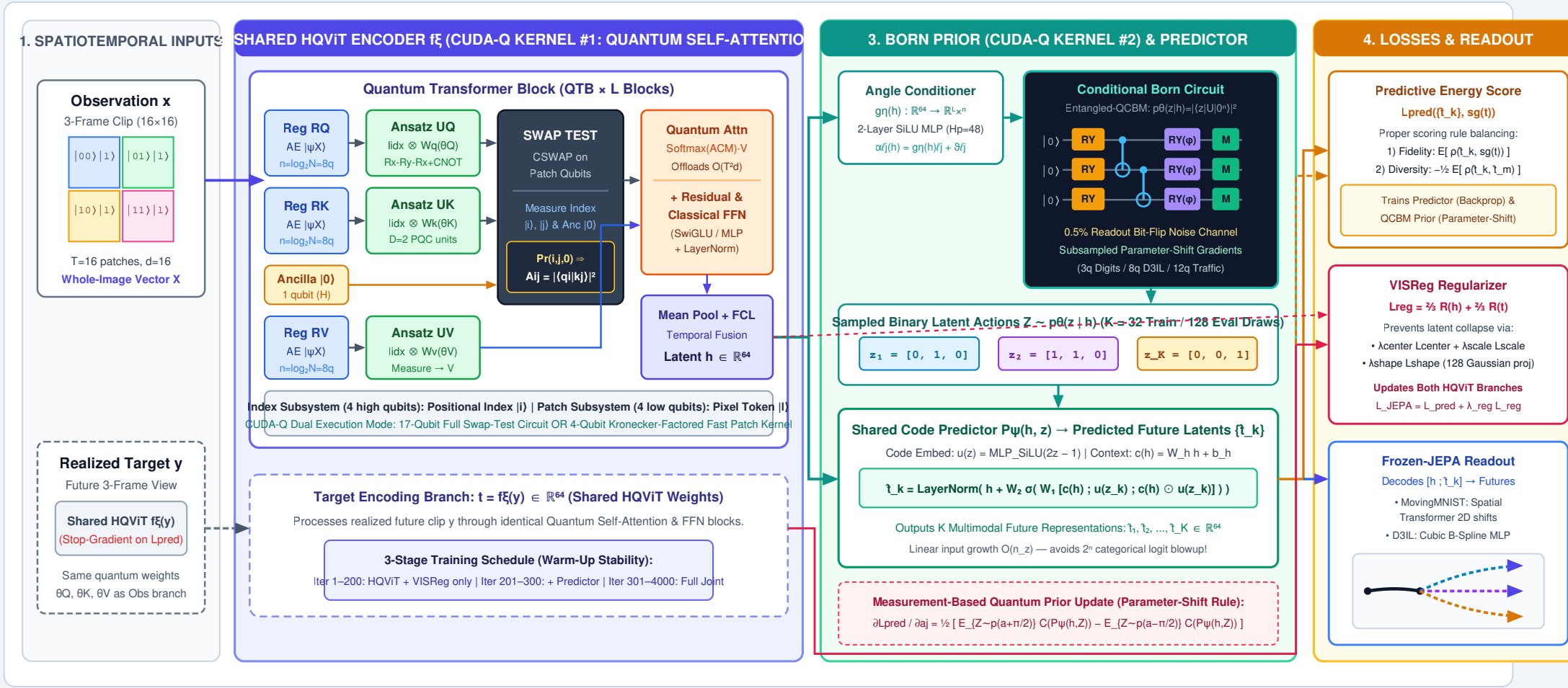
Q²-JEPA (QuViT-JEPA): Dual-Quantum Visual World Model Architecture

Unifying HQViT Hilbert-Space Self-Attention & Conditional Quantum Born Priors for Multimodal Spatiotemporal Prediction

HPC + AI + Quantum | NVIDIA CUDA-Q



1. Quantum Perception (HQViT Encoder)

- **Whole-Image Amplitude Encoding:** Encodes $N=Td$ pixels into $n = \log_2(Td)$ qubits ($8q$ for 16×16), preserving global token positions in the Index Subsystem without extra PE.
- **Swap-Test Self-Attention:** Offloads $O(T^2d)$ attention matrix calculation via $A_{ij} = |\langle q_i|k_j\rangle|^2$ with only $O(\log_2 d)$ parameterized gates.

2. Quantum Uncertainty (Born Prior)

- **Conditional Entangled-QCBM:** Maps observation latent $h \in \mathbb{R}^{64}$ via SiLU conditioner $g_\eta(h)$ to circuit angles $\alpha_{\ell j}(h)$, sampling binary codes $z \in \{0,1\}^n$.
- **Compact Correlated Actions:** CNOT entanglement captures cross-bit correlations with $O(nL)$ gates, avoiding 2^n Softmax-Categorical parameter blowup.

3. Sampled Energy & VISReg Training

- **Leave-One-Out Energy Score:** Trains shared predictor $P_\psi(h,z)$ and QCBM prior directly from K sampled futures without requiring explicit prior likelihoods.
- **Collapse Prevention:** Decoupled VISReg ($L_{center}, L_{scale}, L_{shape}$) regularizes both h and t across a 3-stage warm-up schedule.

4. NVIDIA CUDA-Q HPC Acceleration

- **Kronecker-Factored vs. Full Swap-Test:** Benchmarks full $17q$ Swap-Test simulation against mathematically exact $4q$ batched patch evolution on GPU.
- **Noisy NISQ Emulation:** Evaluates subsampled parameter-shift gradients (r/G angles, b/B batch) under 0.5% readout bit-flip noise.